New leads for the algebraic IC

New leads for the algebraic IC: prior art, a cycle stop, torsion anchors

Status: exploratory. No frozen protocol, no replay certificate, no validator report, so nothing here is a leaderboard or benchmarks row. Producer: `examples/ic_m2_cycle.rs` (branch `claude/ic-m2-cycle-stop-20261029`). Raw records: `raw/`. One thread of a shared 4-vCPU cloud container; the n = 41 run overlapped a d-sweep, so its same-process rho time is noisy (about 10 ms).

Summary

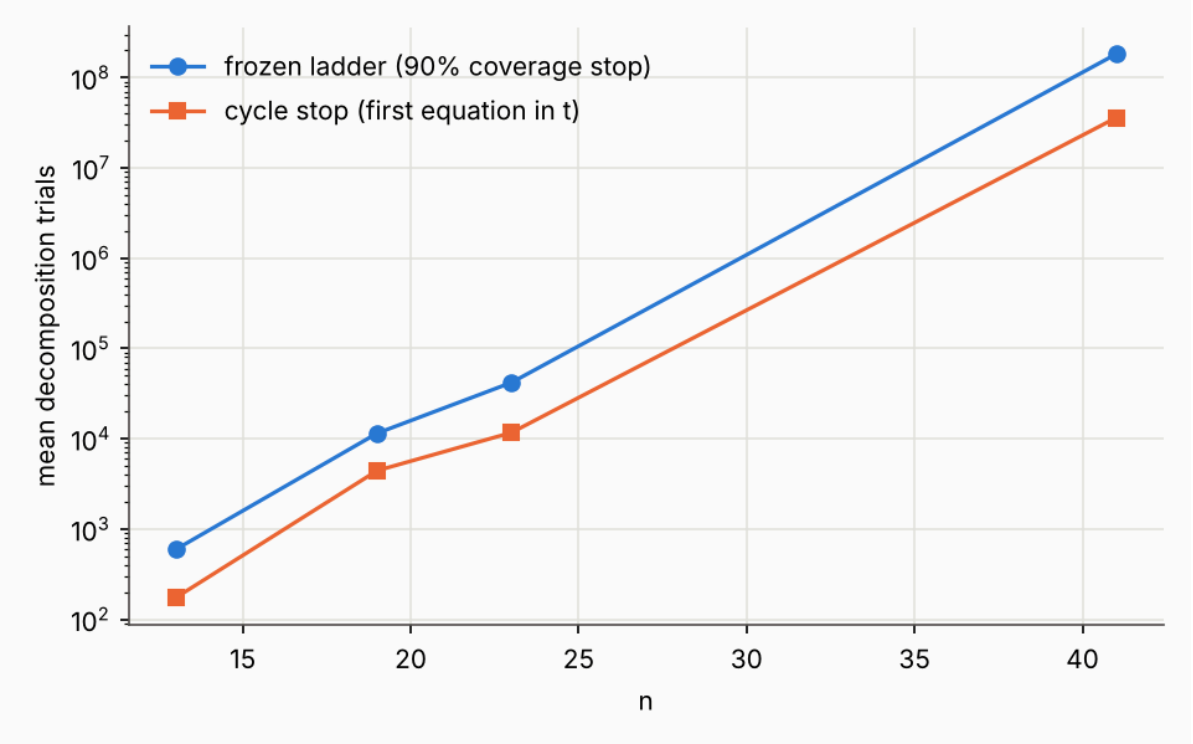
1. **The landed ladder is not new algebra.** Two repository sweeps and a literature pass found the same construction earlier (below). The packaging (symmetric `s, p`, the no-reduction window `2d − 1 ≤ n`, the `2^{3d−1−n}` floor) was not found as a stated theorem, so it is at most an exposition.
2. **A cycle stop is a constant-factor gain.** 5.1× fewer trials and 5.3× less wall time at n = 41; 2.6–3.6× at n = 13–23. Same exponent.
3. **Torsion-randomized anchors make large-cofactor curves workable.** a = 0, n = 37 (cofactor 596) was stopped unfinished by the ladder after 99 s; it is solved here in 26 and 31 s. They *cost* 1.5× (n = 23) to 16× (n = 31) on cofactor-4 curves, so they are for large cofactors only.
4. **The wall-time optimum stays at d ≈ n/3.** Trials keep falling for larger d, but the per-trial solution enumeration grows as `2^{3d−1−n}`.
5. **Beating rho is closed for this family** if the ceiling argument below holds. The distance is about 9,700× at n = 41.

Measurements

Same seed and secrets as the frozen ladder (`ic_m2_linear`, seed 20261024), so rows pair by target. Every row verified (`[t]G = T`); the counters `ic_bad_t` and `ic_inconsistent` were 0 on all rows.

a	n	d	targets	ladder mean trials	cycle mean trials	ratio
0	13	4	8	592	172	3.4
0	19	6	8	11,477	4,427	2.6
1	19	6	8	12,077	3,961	3.0
0	23	8	8	41,856	11,678	3.6
1	23	8	8	48,821	18,521	2.6
0	41	14	4	181,803,464	35,847,188	5.1

n = 41 wall: 478.4 s (ladder) → 90.1 s (cycle), 5.3×. Per-target trial ratios there are 7.2, 3.1, 2.9 and 50.7 (the last target stopped very early), so the mean ratio is driven by four draws; treat the 5× as a point estimate with a wide interval. Against this run's same-process strong rho (9.3 ms mean) the cycle variant is 9,700× slower; the ladder's own record gave 36,440× because its rho draws were slower (13.1 ms), and rho times at this scale are noisy.



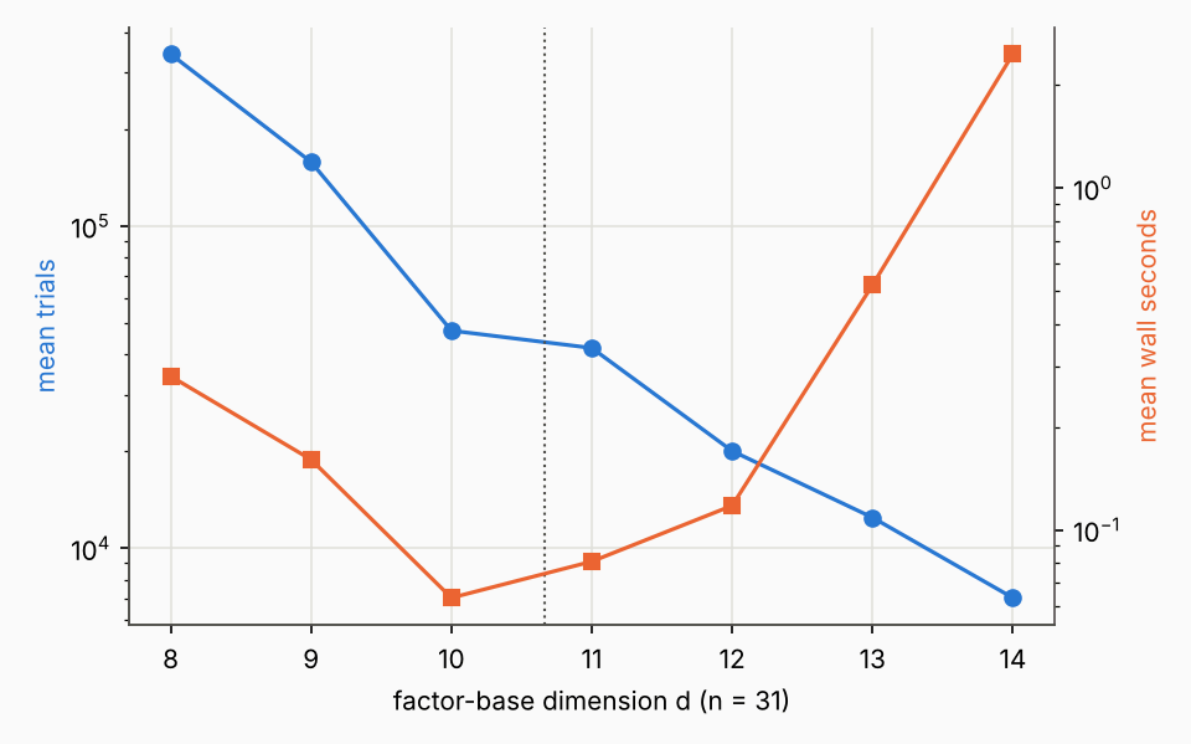
trials against n

Torsion anchors (`raw/cyc_tors_*`, `raw/cyc_a0_n31_t*`, `raw/cyc_a0_n23_t*`):

curve	cofactor	torsion	mean trials	wall	verified
a = 0, n = 37	596	on	16.4 M	28 s	2/2
a = 0, n = 31	4	off	47,411	0.06 s	4/4
a = 0, n = 31	4	on	744,299	0.97 s	4/4
a = 0, n = 23	4	off	11,678		8/8
a = 0, n = 23	4	on	17,062		8/8

The n = 37 run is two targets with no torsion-off comparison (the ladder could not finish it), so the claim is “solvable”, not a measured speedup.

d-sweep at n = 31 (`raw/dsweep_*`, 4 targets each): trials 342k, 158k, 47k, 42k, 20k, 12k, 7k for d = 8…14; wall 0.28, 0.16, **0.063**, 0.080, 0.117, 0.52, 2.46 s. Minimum at d = 10 ≈ n/3.



d sweep

Prior art (what not to re-propose)

Repository paths are relative to `research/` of the `crypto` repository.

- Linear (A, P) oracle over a geometric V, n = 19–31: `geometric_v_linear_20261006`.
- Union-find, no linear algebra, n = 13–23, projected 2^85.7 at n = 131: `graph_cycle_ic_20261007` (its walk found a rho collision first in 20/20 runs).
- Same idea in the cryptanalysis repository: `experiments/linearized-half-decomposition`.
- Literature (read this session by a survey agent, claims marked there as verified from the PDF or from the abstract): Faugère–Perret–Petit–Renault 2012 (subspace base, linearization, n = 41 and 131 tables), Petit–Quisquater 2012, Kosters–Yeo 2015 (the S3 trace morphism gives first fall degree 2), Karabina and Semaev 2015 (chained S3), Galbraith–Gebregiyorgis 2014, Caminata–Ceria–Gorla 2023 (d_reg grows linearly in n).

Ceiling argument (from the repo, not re-derived here): `decomposition_programme_20261016/PLAN.md` states that for prime n, `dim(V·V) ≥ 2l − 1` (Hou–Leung–Xiang), so any pair test over a subspace base needs about `2^{n−l}` tests and loses to rho unless `l > n/2`, which that bound rules out. Even a free pair solver would only reach rho parity. I have not independently checked this bound.

My own arithmetic check: a naive hybrid (guess `g` variables, linearize the rest of the `2d`-variable quadratic system in `n` equations) costs about `2^{2d − √(2n)}`, worse than the `2^{3d−1−n}` enumeration whenever `d < n + 1 − √(2n)`, so it does not help in the range of interest.

Leads, ranked

#	lead	status	why
1	Torsion anchors for large cofactors (N83 a = 1, h ≈	**tested at h = 596; not at N83**	Only lead the sweeps found untried; first positive result.

#	lead	status	why
	$1.1 \cdot 10^9$)		Next: n = 41 and 43-ish rungs with $h \geq 10^3$, torsion on/off.
2	Mixed-conjugate excess `dim(V·V^{2^k}) − (2l−1)` (W2)	proposed, not run	Could add free relations without changing the base.
3	4–5-summand test inside a frozen budget (W3)	not run	Uses `S4 = G²` as one condition. Prior 13/40 → 0/40 full rank at l = 5, 6 (n = 131) limits reach.
4	Crossbred / last-fall-degree on Weil-descended S systems	never beyond n = 9	Cheap instruments; end-to-end enumeration beat crossbred 3.4–1800× at n ≤ 13.
5	Splitting predicate as bilinear/MinRank for d in (n/3, n/2)	open, low ceiling	Best case approaches rho parity, not below it.
6	Decisive Weil-descent cell (m, l) = (4, 6)	blocked on about 70 GB	Memory, not ideas.

Caution on n = 41: the survey agent notes `ord_41(2) = 20`, so F_{2^41} has 20-dimensional Frobenius-invariant subspaces and n = 41 is unrepresentative of n = 83 (`ord_83(2) = 82`) and n = 131.

What this does not show

- No exponent change: the cycle stop is a constant factor (fit not rerun).
- Four targets at n = 41 and eight below; one container; no validator.
- The torsion result is one curve, two targets.
- Nothing extrapolates to N83. On a = 0 the projected cost is still about 2^55 trials against 2^37 rho steps (from the landed report); on a = 1 torsion anchors are untested.